# Characterisation and Quantification of Data Centre Flexibility for Power System Support

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**Highlights**

- Linked annual optimisation preserves physical and workload continuity.
- Integrated operation cuts annual electricity cost by 5.049%.
- Flexible workload has greater value than UPS or TES capacity.
- A 25 kW import increase is sustained for 9.75 h on the representative day.
- Component decomposition identifies how each grid response is delivered.

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***Abstract*—The rapid growth of the data centre (DC) sector creates both a power-system challenge and an opportunity for demand-side flexibility. This paper develops a linked rolling-horizon mixed-integer optimisation of IT workload scheduling, uninterruptible power supply (UPS) dispatch, cooling dynamics and thermal energy storage (TES) for a 1 MW DC. Physical states and unfinished workload cohorts are carried exactly between local-day horizons across a full year, including daylight-saving transitions. Against a workload-at-arrival, storage-disabled benchmark with cost-optimised cooling, coordinated operation reduces 2025 electricity cost from GBP 550,503.59 to GBP 522,707.23, a saving of 5.049%. Full-year endpoints rank flexible workload as the most influential tested resource, followed by UPS and TES energy capacity. An objectively selected representative day is then subjected to 288 grid-deviation requests. Each request begins from the stored annual state, follows the annual pre-event CPU trajectory and must recover storage and thermal states after the event. The resulting duration envelope is strongly time- and direction-dependent: increased import is sustained for up to 9.75 hours at +25 kW, whereas reduced import is sustained for up to 5.25 hours at -100 and -150 kW. An additive component decomposition identifies the contributions of IT load, CRAC cooling, TES charging and UPS dispatch. All continuity, workload, exclusivity, settlement and flexibility-response audits pass. Of 288 cells, 283 have exact duration boundaries and five retain explicitly marked certified lower bounds, establishing a reproducible link between annual economic operation and recovery-certified grid flexibility.**

***Index Terms*—Data Centre, Power system flexibility, Demand side flexibility, Load Shifting, Thermal Inertia**

## I. Introduction

GLOBAL digitalisation, propelled by artificial intelligence, is driving the rapid expansion of DCs, whose electricity demand is both substantial and spatially concentrated. This growth coincides with two other major shifts in the energy landscape: the increasing share of variable renewable generation needed for decarbonisation, and the widespread adoption of new demand-side technologies such as electric vehicles and heat pumps [1], [2]. The combined effect places significant strain on modern power systems by intensifying peak loads, steepening net load ramps, and aggravating local network constraints. Consequently, there is an escalating need for power system flexibility to maintain the crucial balance between electricity supply and demand.

According to the IEA, global DC electricity consumption was around 415 TWh in 2024, with projections suggesting this will exceed 945 TWh by 2030, raising its share of global electricity demand from approximately 1.5% to 3% [3]. This expansion is not uniform, with over 80% of the increase expected to originate from the United States and China [3]. Moreover, geographical clustering intensifies the impact on local grids. For instance, DCs consume around 20% of Ireland's metered electricity supply, while their share in Virginia, USA, is 25% [3]. In Great Britain, annual electricity consumption by DCs is forecast by National Grid ESO to rise from 7.6 TWh in 2024 to between 30–71 TWh by 2050 [4].

To manage the volatility introduced by these trends, power systems require a substantial increase in flexibility services. Traditionally provided by dispatchable fossil fuel generators, this solution is becoming obsolete under decarbonisation mandates. The JRC [5] projects that the European Union's flexibility requirement will rise to 24% of total electricity demand in TWh by 2030 and to 30% by 2050. Similarly, Great Britain's Clean Flexibility Roadmap gives a 2030 clean-flexibility capacity range of 51–66 GW, compared with 25.2 GW in 2024 [6]. Demand-side flexibility, where consumers actively modulate their electricity use to support the grid, presents a key solution.

While their growth contributes to the challenge, DCs are equipped to provide flexibility. Their shiftable IT workloads, uninterruptible power supply (UPS) systems with battery storage, and inherent thermal inertia as well as cold storage, can be leveraged to modulate power consumption without disrupting core services. Each of these internal assets possesses a distinct flexibility profile in terms of magnitude, duration, ramp rate, and recovery time [7]. For example, UPS batteries can offer a near instantaneous response, whereas thermal systems are characterised by lower ramp rates, providing a more gradual adjustment [7], [8]. DCs therefore present a dichotomy: they are both a growing strain on the power system and a potentially crucial asset for grid flexibility.

In the context of power systems, flexibility is defined as the capability of maintaining the balance between generation and demand across all time horizons, by adjusting either energy production or consumption in response to sudden variations [1], [2], [4], [9]. Upward flexibility is required when demand exceeds generation and involves a DC reducing its power consumption from the grid. Conversely, downward flexibility is needed when generation surpasses demand, and is provided by a DC increasing its electricity consumption to absorb the surplus energy [1], [10]. In this study, a DC's flexibility is quantified by its capacity for power consumption deviation relative to a baseline, and the maximum duration for which this deviation can be sustained.

This paper addresses these challenges with a linked annual optimisation of IT workload, UPS dispatch and cooling operation against signed electricity prices. Its principal contribution is an auditable rolling-horizon formulation that passes physical states and unfinished workload between local days instead of resetting the facility at midnight. The study quantifies annual cost value, tests full-year sensitivity endpoints for the three internal flexibility resources, and defines a recovery-constrained duration assessment to be applied to an objectively selected representative day.

The main contributions are therefore: (1) a tractable integrated data-centre model that preserves the relevant coupling between IT execution, UPS operation, cooling power and TES state; (2) a complete 2025 rolling chain with explicit state, workload, solver and settlement audits; and (3) five-point full-year sensitivity evidence for flexible workload, UPS capacity and TES capacity.

The remainder of this paper reviews the literature, defines the linked annual method and benchmark, presents the completed annual results and sensitivities, and identifies the deferred representative-day flexibility assessment.

### A. DC Architecture

DCs are integrated facilities housing IT equipment, such as servers and network devices, which are supported by critical power and cooling infrastructure. As shown in Figure 1, electrical continuity is maintained by uninterruptible power supplies (UPS) with associated battery systems, while cooling systems, including chillers, Thermal Energy Storage (TES) tanks and Computer Room Air Conditioning (CRAC) units, manage the substantial heat generated by the servers. Thermal efficiency is often enhanced through optimised airflow strategies like the hot-aisle/cold-aisle arrangement.

Each of these core subsystems presents an opportunity for providing demand-side flexibility to the power grid. The strategic scheduling of IT workloads, the dispatch of UPS battery storage, and the leveraging of the TES tank and the facility's inherent thermal inertia allow a DC to modulate its power consumption. The DC assets are highly interdependent, requiring an integrated approach that considers all of them simultaneously. This is because a change in one subsystem changes the feasible operating range of the others: deferring IT workload reduces both server power and subsequent heat generation, cooling actions are constrained by thermal states and the TES charge level, and UPS discharge can support IT execution while altering the grid power draw. A component-by-component assessment can therefore overestimate flexibility if it ignores recovery requirements, or underestimate it if it misses compensating actions across assets. Consequently, DCs can transition from being passive electrical loads to active assets that enhance the resilience and flexibility of the power system.

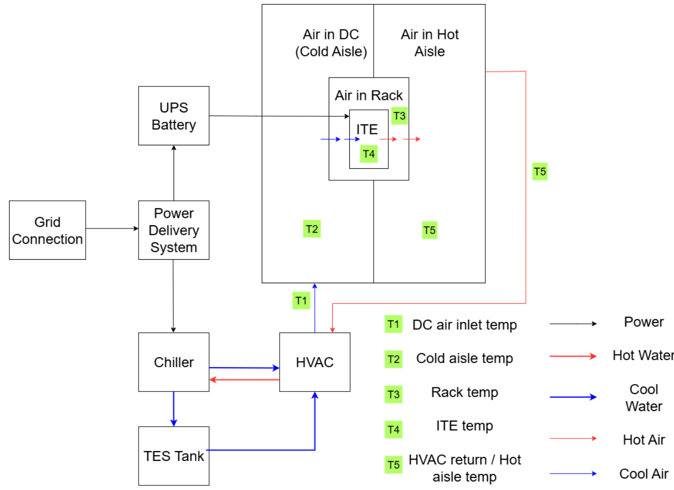



Fig. 1. DC Layout

### B. IT Workload Background

The fundamental role of a DC is to execute IT workloads, which encompass all computational and data processing tasks. These workloads can be broadly classified based on their time sensitivity, as defined by service-level agreements (SLAs) [11], [12]. Inflexible workloads, also known as interactive workloads, are latency-critical and require immediate execution to ensure quality of service for user-facing applications like real-time streaming or transaction processing.

In contrast, flexible IT workloads, often referred to as batch workloads, can tolerate execution delays within predefined time windows without impacting service quality. These typically non-user-facing tasks, such as periodic data analytics, backups, or machine learning model training, can be deferred or rescheduled. This ability to shift the timing of their execution is the primary source of IT-based flexibility, allowing the DC to adjust its computational power draw in response to external signals like electricity prices or grid operational needs [12].

## II. Literature Review

DC flexibility has attracted growing attention as an enabler of demand-side participation in power systems, with a growing body of literature modelling flexibility at the asset level, predominantly focusing on IT workloads and, to a lesser extent, UPS batteries, cooling systems and other subsystems. A smaller body of work has attempted integrated models that capture the joint flexibility of multiple DC assets. This review therefore first presents the flexibility modelling approaches found in the literature for each DC asset, and then discusses studies that move towards an integrated DC flexibility assessment.

IT workload is the most fundamental component of a DC and is the focus for many flexibility studies. One study develops a machine learning method to predict DC energy consumption and server temperature [13]. These data are then used to optimise the scheduling of IT workloads to minimise fuel costs and power utilisation at times of high grid carbon intensity. On average, the implementation achieved a 6.5% load reduction for a winter day during periods of high $CO_2$ intensity. Radovanović et al. [14], performed predictive modelling of grid carbon intensity, trained day-ahead demand prediction models and used them to optimise IT workload for minimum carbon emissions. Cao et al. [12] propose and validate a data-driven methodology using real-world Alibaba trace data to assess the power flexibility potential of Internet Data Centres (IDCs) by rescheduling periodic IT workloads. Their findings quantify significant flexibility, particularly in early morning hours, revealing potential upward/downward load shifts of 400 kW and 325 kW, respectively—corresponding to approximately 50% and 40.6% of the facility's peak demand—by optimising job schedules based on power system needs and renewable energy availability. In this study, IT workloads are defined as either flexible or inflexible, with flexibility here referring to flexibility in time.

UPS batteries are currently providing power flexibility to the grid in numerous DCs around the world and have been shown to be successful [15]–[17]. Lithium-ion batteries, which are increasingly the dominant form used for UPS, are capable of rapidly providing or consuming power when needed [8]. These characteristics make them well suited to frequency regulation, a service many power providers require. Results from [8] found UPS batteries could be used within their normal operating range to provide the required services to the grid and in doing so generate a reasonable revenue stream. Furthermore, the battery lifetime would not be affected, and the approach does not require additional cost.

Cooling makes up a large proportion, as high as 40%, of DC energy consumption [18]. The American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) recommends maintaining an indoor temperature of $18°C$ to $27°C$ to ensure optimal DC performance [19]. Exploiting the thermal inertia provided by this temperature range has been shown to enable power consumption flexibility in DCs [20] [21]. Thermal Energy Storage (TES) systems can enhance cooling flexibility by decoupling the timing of heat removal from the cooling generation process. Du et al. [22] propose and evaluate a framework to enhance the energy flexibility of district heating (DH) systems integrated with DC waste heat recovery. The study utilises dual short-term TES tanks to enable simultaneous peak shaving (demand-driven) and load shifting (price-driven) within the hybrid system. Simulation results demonstrated that this dual TES approach significantly improved flexibility, achieving up to a 10% peak demand reduction and a 2.1% load shift, leading to a 3.2% operational cost saving in the case study. A synergistic control strategy for DC frequency regulation using both IT and cooling systems has also been proposed [23]. Results found that a revenue saving of 4% could be made by implementing the proposed strategy. Once depleted, the thermal inertia of the DC atmosphere / TES must be recovered to baseline levels. The recovery time taken to do so will affect how the flexibility of the asset can be utilised.

While individual assets offer flexibility, the flexibility characteristics change when these assets are integrated into one system. Maximising the overall potential requires coordinating multiple resources and evaluating their diverse characteristics, interactions, and constraints. One study proposes a co-optimisation of the IT workload and cooling system, recognising the strong operational link between them [23]. The work by S Xiang et al. [24] identifies that DC optimisation models often ignore IT equipment constraints like start-stop conditions and ramp rates, which can cause excessive wear. The authors propose a model integrating these constraints with facility thermal dynamics, solved using a Deep Deterministic Policy Gradient (DDPG) algorithm. This approach was then validated for practicality and efficiency using a large-scale simulation of a 100,000-device DC. H. Xu et al. [25] propose an optimisation which integrates IT servers, cooling infrastructure, energy storage, generators and renewable energy, and aims to maximise power demand flexibility. The objective function is the cumulative change in energy taken from the grid before and after the optimisation was applied. Other studies use a similar approach, but instead minimise the difference between DC power consumption and the desired power consumption of the power supply operator [26].

While individual DC assets, including IT workloads, UPS batteries and cooling systems, have each been studied for their flexibility potential, the interplay between these assets and its impact on the overall flexibility of the DC remains largely unexplored. Most existing studies adopt a DC-centric, cost-minimisation perspective and do not quantify the magnitude and duration of grid-facing flexibility that can be delivered once coupled asset constraints and post-event recovery are accounted for. As DCs scale towards the gigawatt level, this power-system-centric perspective becomes increasingly important, because system operators require duration-certified flexibility offers and a clear understanding of which assets enable them; this perspective remains largely absent from the literature.

Addressing these gaps, this study makes three contributions. First, it establishes an audited full-year operating baseline in which physical states and deadline-constrained work are linked between daily horizons. Second, it quantifies how workload flexibility, UPS capacity and TES capacity affect the annual economic result using completed full-year endpoints. Third, it maps the maximum recovery-certified duration of grid-power deviations on an objectively selected day and decomposes each response into its IT, cooling, TES and UPS contributions. The annual and event studies use the same stored trajectory, so the reported service envelope is conditional on an internally consistent cost-optimal operating state rather than an independently reset day.

## III. Methodology

This section details the mathematical framework developed to quantify the demand-side flexibility of a DC by co-optimising its primary subsystems. The conceptual DC model, illustrated in Figure 1, integrates three dynamically coupled components: the IT systems, which process both flexible and inflexible computational workloads; the power infrastructure, featuring an Uninterruptible Power Supply which operates as an Energy Storage System (UPS-ESS); and the cooling infrastructure, which includes a chiller coupled with a Thermal Energy Storage (TES) tank and the CRAC system.

Our approach is centred on a comprehensive optimisation model that captures the operational dynamics and physical constraints of each subsystem. Flexibility is harnessed from three key sources: (1) shifting IT workloads in time, (2) strategically charging and discharging the UPS-ESS, and (3) strategically managing the cooling system by utilising the TES and the inherent thermal inertia of the facility. By integrating these components into a unified framework, the DC's total capacity to modulate its electricity consumption is evaluated, thereby quantifying its potential to provide valuable grid services.

The model is intentionally aggregate rather than a server-level or device-level digital twin. The objective of this work is to quantify the flexibility potential available to the power system from a DC, not to reproduce the detailed operational behaviour of individual servers or cooling devices [27], [28]. The modelling therefore prioritises the state variables that determine flexibility feasibility: CPU utilisation and delay tolerance for IT work, UPS state of charge and converter limits for electrical storage, TES state of charge and facility temperatures for cooling recovery, and total grid power for service delivery. Likewise, the purpose of the cooling model is to capture the energy-flexibility interaction between thermal storage, cooling demand and electricity consumption, rather than detailed airflow dynamics within server racks. This abstraction preserves the operational couplings that matter for flexibility while keeping the repeated optimisation solves required by the flexibility assessment computationally tractable. Importantly, the required computational work of every IT job is always executed in full within its delay tolerance, so flexibility is provided without sacrificing computational throughput or service quality.

The revised implementation applies this integrated model across the complete 2025 calendar year. Rather than solving 365 independent days, it links the physical state and unfinished computational work from one local day to the next. Each local-day optimisation contains a committed core and a three-hour look ahead, and the detailed annual handoff procedure is given at the end of this section. The component models are first presented in the same order as the original formulation so that the role and revision of each equation remain transparent.

Table I
Evidence Used to Construct the Workload Assumptions

| Evidence type | Sources | Use in this study |
|---|---|---|
| Observed utilisation ranges | [29], [30] | Bounds and representative average CPU utilisation levels. |
| Delay-sensitivity classes | [31] | Mapping of workload classes into flexible and inflexible groups. |
| Batch-workload flexibility | [12] | Flexible/inflexible workload energy shares and periodic workload shifting behaviour. |
| Flexibility case studies | [13], [23], [25], [32], [33] | Cross-checking plausible daily workload shapes and flexibility ranges. |

### A. IT Workload Methodology

In the literature, there is no consensus on IT workload characteristics and utilisation levels, as they vary significantly depending on DC size, type (e.g., enterprise, colocation, hyperscale) and workload nature (e.g., banking, AI). Consequently, even publicly available real-world datasets can differ substantially across facilities. Chris Zaloumis [29] indicates that average CPU utilisations vary between 12% and 18% of total capacity, while Ankur Ghia's findings indicate a daily average between 20% and 30%, noting that best practices can elevate this to 70%–80% [30].

A study by Google [31] classified DC workloads into four categories based on their sensitivity to delay, labelled from "0" (least sensitive) to "3" (most sensitive). If categories 2 and 3 are considered inflexible, they account for approximately 30% of the total workloads, suggesting that up to 70% of the workloads could be flexible. Another study by Cao et al. [12], conducted using Alibaba's cluster trace, found that inflexible workloads represent 60% of all jobs but contribute only 30% of total energy consumption. In contrast, flexible workloads constitute 40% of jobs while consuming 70% of the energy. Furthermore, studies such as [13], [23], [25], [32], [33] reveal that there is no universally accepted ratio, and the estimated shares of flexible and inflexible IT workloads differ significantly across studies for a typical 24-hour period.

Based on the review of the literature, a dataset of reported workload ratios and graphs over a 24-hour time horizon was compiled. From this dataset, the minimum and maximum values were extracted, and the average ratio was computed, separately for flexible, inflexible, and total workloads relative to total CPU capacity. Drawing upon these values, the workload ratio assumptions adopted in this study were established and are presented under the "Proposed" category in Table II. The evidence base used to construct these assumptions is consolidated in Table I. Reported hourly workload ratios and graphical profiles were tabulated or digitised where required and grouped into flexible, inflexible and total workload categories. For each hour, the minimum, maximum and average reported values were computed, and the proposed value was selected as a representative profile within the observed range while preserving the daily shape and the aggregate balance between flexible and inflexible demand. The "Proposed" columns therefore constitute modelling assumptions synthesised from the literature rather than measurements from a single facility.

Categorising the deferral flexibility of IT workloads, defined

Table II
IT WORKLOAD RATES

| Time | Flexible Workload Ratio (%) | | | | Inflexible Workload Ratio (%) | | | | Total Workload Ratio (%) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Min | Max | Average | Proposed | Min | Max | Average | Proposed | Min | Max | Average | Proposed |
| 00–01 | 17 | 75 | 37 | 40 | 20 | 39 | 32 | 28 | 55 | 80 | 68 | 68 |
| 01–02 | 13 | 72 | 36 | 31 | 25 | 38 | 32 | 25 | 46 | 66 | 56 | 56 |
| 02–03 | 18 | 70 | 38 | 33 | 25 | 37 | 32 | 17 | 43 | 50 | 47 | 50 |
| 03–04 | 15 | 68 | 37 | 23 | 19 | 38 | 26 | 16 | 34 | 37 | 36 | 39 |
| 04–05 | 17 | 64 | 36 | 27 | 8 | 36 | 21 | 8 | 25 | 37 | 31 | 35 |
| 05–06 | 15 | 59 | 32 | 27 | 2 | 39 | 21 | 6 | 22 | 36 | 29 | 33 |
| 06–07 | 14 | 51 | 29 | 18 | 13 | 40 | 25 | 12 | 20 | 36 | 28 | 30 |
| 07–08 | 24 | 42 | 31 | 24 | 18 | 39 | 27 | 20 | 24 | 52 | 38 | 44 |
| 08–09 | 23 | 35 | 30 | 24 | 30 | 53 | 41 | 24 | 31 | 65 | 48 | 48 |
| 09–10 | 27 | 49 | 35 | 28 | 36 | 65 | 49 | 34 | 38 | 85 | 62 | 62 |
| 10–11 | 23 | 50 | 35 | 19 | 37 | 62 | 51 | 37 | 45 | 87 | 66 | 56 |
| 11–12 | 19 | 50 | 35 | 18 | 42 | 70 | 56 | 42 | 50 | 92 | 71 | 60 |
| 12–13 | 17 | 49 | 33 | 20 | 40 | 51 | 47 | 40 | 52 | 89 | 71 | 60 |
| 13–14 | 16 | 49 | 33 | 24 | 36 | 60 | 49 | 36 | 54 | 85 | 70 | 60 |
| 14–15 | 16 | 52 | 35 | 27 | 35 | 68 | 52 | 35 | 55 | 87 | 71 | 62 |
| 15–16 | 17 | 49 | 35 | 27 | 33 | 70 | 53 | 33 | 57 | 82 | 70 | 60 |
| 16–17 | 26 | 43 | 37 | 20 | 35 | 62 | 52 | 40 | 63 | 78 | 71 | 60 |
| 17–18 | 37 | 39 | 38 | 40 | 20 | 60 | 39 | 27 | 67 | 75 | 71 | 67 |
| 18–19 | 32 | 49 | 39 | 45 | 10 | 63 | 37 | 26 | 70 | 73 | 72 | 71 |
| 19–20 | 23 | 54 | 35 | 45 | 9 | 67 | 38 | 26 | 62 | 77 | 70 | 71 |
| 20–21 | 22 | 58 | 36 | 47 | 8 | 62 | 37 | 25 | 63 | 83 | 73 | 72 |
| 21–22 | 21 | 61 | 38 | 41 | 7 | 61 | 37 | 29 | 63 | 87 | 75 | 70 |
| 22–23 | 17 | 64 | 38 | 40 | 6 | 55 | 35 | 30 | 61 | 89 | 75 | 70 |
| 23–24 | 16 | 69 | 40 | 42 | 2 | 45 | 29 | 21 | 56 | 80 | 68 | 63 |

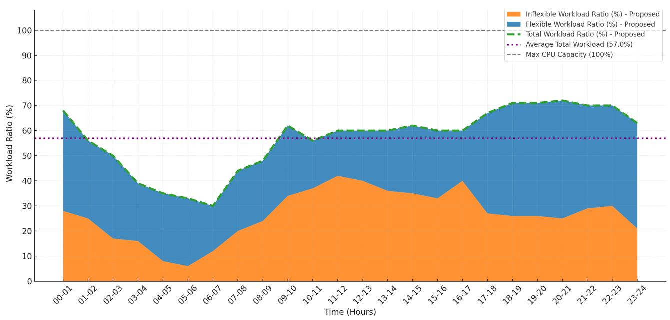



Fig. 2. Stacked Proposed Workload Ratios Over 24 Hours

Table III
HOURLY DISTRIBUTION OF SHIFTABLE WORKLOAD UNDER DIFFERENT DEFERRAL WINDOWS (%)

| Tranche | Deferral | Time | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 00 | 01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 |
| $k = 1$ | $0 \leq 30$ min | 25 | 25 | 35 | 25 | 25 | 25 | 23 | 20 | 25 | 25 | 35 | 32 | 45 | 50 | 40 | 45 | 50 | 50 | 10 | 15 | 18 | 22 | 20 | 15 |
| $k = 2$ | $0 \leq 60$ min | 25 | 25 | 17 | 15 | 20 | 15 | 28 | 20 | 15 | 17 | 22 | 20 | 25 | 20 | 20 | 25 | 15 | 20 | 20 | 20 | 12 | 18 | 16 | 20 |
| $k = 3$ | $0 \leq 2$ h | 20 | 15 | 18 | 20 | 15 | 15 | 15 | 15 | 13 | 16 | 13 | 15 | 15 | 15 | 25 | 15 | 20 | 15 | 30 | 25 | 25 | 25 | 24 | 20 |
| $k = 4$ | $0 \leq 3$ h | 30 | 35 | 30 | 40 | 40 | 45 | 34 | 45 | 47 | 42 | 30 | 33 | 15 | 15 | 15 | 15 | 15 | 15 | 40 | 40 | 45 | 35 | 40 | 45 |

as the proportion of workloads that can be shifted and the maximum duration by which they can be postponed, is essential for developing effective load-shifting strategies. Drawing on insights from various studies [1], [12], a representative set of deferral windows applicable to flexible IT workloads over a 24-hour period is defined and summarised in Table III.

Table III presents the hourly distribution of flexible workloads across the four delay tranches: tranche $k = 1$ contains jobs that can be deferred by up to 30 minutes, $k = 2$ by up to 60 minutes, $k = 3$ by up to 2 hours, and $k = 4$ by up to 3 hours. The percentages represent the relative distribution within the flexible workload. For instance, during the 12:00–13:00 interval, 45% of the flexible workloads can be deferred up to 30 minutes, 25% for up to 60 minutes, 15% for up to 2 hours, and the remaining 15% for up to 3 hours. These categorized hourly workload profiles are shown in Figure 2 and utilised as modelling inputs, capturing the time-dependent and duration-sensitive characteristics of DC flexibility in this study.

*1) IT Power Consumption Model:* In the literature, various approaches exist for modelling the power consumption of IT equipment, particularly servers. Given that server power consumption often constitutes the dominant portion of the total IT power draw, other components are frequently disregarded, and server power consumption is typically taken as a proxy for the overall IT power consumption. Numerous linear and non-linear models have been developed to represent server power consumption. A fundamental and widely accepted principle is that server power consumption is strongly correlated with its CPU utilisation. Based on this principle, Equation 1 was used in this study to estimate the electric power consumption associated with the IT workload [28], [34].

$$P^{\mathrm{IT}}(t) = P^{\mathrm{IT}}_{\mathrm{idle}} + (P^{\mathrm{IT}}_{\mathrm{max}} - P^{\mathrm{IT}}_{\mathrm{idle}}) \times u(t)^{1.32} \quad (1)$$

Therefore, the flexibility provided through IT workload shifting fundamentally depends on the rescheduling of CPU utilisation over time.

The core principle of workload shifting employed in this article relies on the concept of constant computational demand for each job. This demand, denoted as $R$, is defined by the required CPU resources and the duration for which they are needed. Representing computational demand as the product of CPU utilisation and execution duration is a common approximation in aggregate workload modelling, because the total amount of computational work can be interpreted as the accumulation of processor occupancy over time [12], [28], [31]. For example, a workload operating at 50% utilisation for two hours produces an equivalent computational demand to a workload operating at 100% utilisation for one hour. This abstraction is appropriate at the facility level considered here, where the modelling target is total processor occupancy over time rather than the instruction-level execution path of individual jobs. Execution can be at a lower or higher CPU utilisation for a longer or shorter duration respectively. In all cases, the IT job must be fully executed within the maximum delay tolerance defined by its tranche. The crucial relationship of constant computational demand for a given IT job is mathematically represented in Equation 2 and conceptually illustrated in Figure 3.

$$R = u(t) \times \Delta T \quad (2)$$

As depicted in Figure 3, a job with a constant computational demand $R$, can be executed over varying time frames $\Delta T$, by adjusting the allocated CPU utilisation $u(t)$. Decreasing CPU utilisation extends the job's duration, while increasing it shortens the duration, all while maintaining the same total computational demand ($R$). Consequently, the power consumption profile corresponding to the CPU utilisation changes accordingly. By strategically increasing or decreasing CPU utilisation in response to system requirements or grid signals, the associated power consumption can be modulated, enabling the provision of upward or downward flexibility, respectively.

*2) IT Workload Formulation:* The annual formulation retains the four workload tranches and the time-varying fractions reported in Table III, but represents every arrival as a timestamped cohort. This change is necessary because work

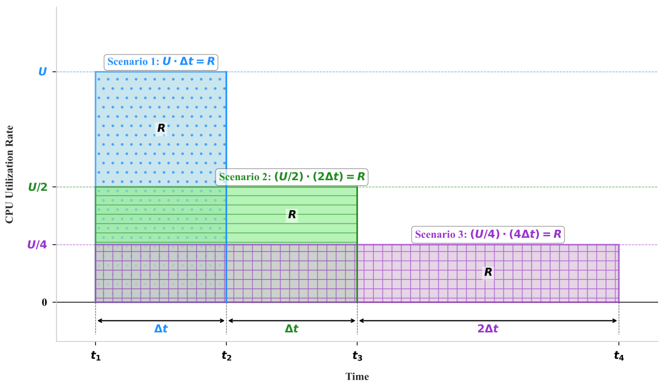



Fig. 3. Constant Computational Demand ($R$) with Varying CPU Utilisation and Duration

arriving near midnight may have a deadline on the following local day. A cohort retains its original arrival time, delay class, absolute deadline and remaining CPU-hours when it is passed between daily optimisations.

The model conceptualises all servers in the DC as one aggregate computational resource. This is the same system-level simplification used in the smaller revision: it does not model placement on individual servers, but it does preserve the amount of computational work, its release time, deadline and the aggregate CPU-capacity limit.

For cohort $c$, let $r_c$ be its arrival time, $d_c$ its absolute deadline and $a_c^d$ the CPU-hours remaining when local day $d$ is opened. The feasible execution window is $\mathcal{W}_c = \{t : r_c \leq t \leq d_c\}$. The decision variable $u_{c,t}$ is the share of aggregate CPU capacity assigned to cohort $c$ in interval $t$. The inflexible workload is executed at arrival, while flexible cohorts may be distributed only within their corresponding window.

If $u_{c,t}$ is the utilisation assigned to the cohort at interval $t$, every cohort whose deadline is visible in the current horizon must satisfy Equation (4). Cohorts with a later deadline may remain partly unfinished, in which case their remaining CPU hours are stored and passed to the following day. At each interval the total scheduled utilisation is limited by Equation (5). Equations (4) and (5) preserve the central idea of constant computational demand: changing the execution rate can change when a job is processed, but not the amount of work that must be completed. This permits load shifting without treating delayed work as an energy saving or a reduction in service quality.

The cohort representation can be understood by considering work that arrives at noon. The part assigned to the 30 minute class may be executed at noon or in either of the following two quarter hour intervals. Work in the three hour class has a wider choice of execution times, but it must still be completed by 15:00. The optimiser may spread either class over several intervals, provided that the area under its utilisation trajectory, measured in CPU hours, equals the original requirement. A lower execution rate requires more time, and a higher rate completes the same work sooner. Equation (5) couples all cohorts that are eligible at the same time and prevents their combined execution from exceeding the available computing capacity.

This treatment separates temporal flexibility from demand destruction. Deferring work can reduce grid demand in one interval, but it creates an obligation to execute the same CPU-hours before the cohort deadline. The load can therefore be shifted, but it cannot be discarded or postponed indefinitely. Similarly, advancing work can increase grid demand during a period when greater consumption is useful, but only if the job has already arrived and computing capacity is available. These restrictions are why the flexible share alone does not determine the response of the whole facility. The location of deadlines, the current CPU loading and the condition of the cooling and storage systems are all relevant.

Aggregate IT electrical power is related to CPU utilisation by Equation (1).

For a cohort that has not reached its deadline by the end of the committed core $\mathcal{C}_d$, the remaining work passed to the next day is calculated using Equation (6). The arrival time and absolute deadline are not changed during this handoff. Consequently, a cohort carried across midnight does not receive a new delay allowance. A cohort whose deadline lies within the current optimisation horizon is subject to Equation (4); a later cohort may remain partly unfinished, but Equation (6) records exactly the obligation that remains.

For clarity, the benchmark and optimised IT quantities can also be written in the form used in the smaller revision, as shown in Equations (7)–(10). In the benchmark, flexible work is executed at arrival and storage dispatch is disabled; it is not reclassified as a different physical workload. In the optimised case, Equation (4) preserves the same CPU-hours while allowing their timing to change. Unlike the earlier extension-only expression, full facility power is modelled in both the committed core and the look-ahead. There is therefore no subtraction of a separate look-ahead baseline.

where $P_{\text{idle}} = 166.7$ kW and $P_{\text{max}} = 1000$ kW [28], [34]. The nonlinear curve is represented in the MILP by a four segment DLOG piecewise linear approximation. Nonuniform breakpoints place more resolution at low utilisation, where the curvature is stronger. Across $0 \leq u \leq 1$, the maximum approximation error is 6.241 kW, or 0.624% of rated IT power.

Equation (1) represents the aggregate servers rather than an individual machine. The idle term captures the power drawn even when no useful work is being executed, while the second term increases with CPU utilisation. Because the exponent is greater than one, the incremental electrical cost of utilisation is not constant across the operating range. Rescheduling work can change both the timing and, to a smaller extent, the electrical energy associated with its execution. The piecewise representation retains this shape while allowing the annual problem to remain a mixed integer linear programme.

### B. UPS ESS Mathematical Model and Constraints

The central UPS has 600 kWh of usable modelled capacity and begins the annual run fully charged. Its minimum permitted state of charge is 50%. Charging and discharging efficiencies are 0.82 and 0.92. With $P_{\text{ch},t}^{\text{UPS}}$ and $P_{\text{dis},t}^{\text{UPS}}$ in kW, the complete UPS formulation is collected below.

$$\mathcal{W}_c = \{t : r_c \le t \le d_c\} \tag{3}$$

$$\sum_{t\in\mathcal{W}_c} u_{c,t}\Delta t = a_c \tag{4}$$

$$u_t = u_t^{\mathrm{inflex}} + \sum_{c:t\in\mathcal{W}_c} u_{c,t} \le 1, \qquad u_{c,t} \ge 0 \tag{5}$$

$$a_c^{d+1} = a_c^d - \sum_{t\in\mathcal{C}_d\cap\mathcal{W}_c} u_{c,t}\Delta t \tag{6}$$

$$u_t^{\mathrm{base}} = u_t^{\mathrm{inflex}} + u_t^{\mathrm{flex,arrival}} \tag{7}$$

$$P_t^{\mathrm{IT,base}} = P_{\mathrm{idle}} + (P_{\mathrm{max}} - P_{\mathrm{idle}})(u_t^{\mathrm{base}})^{1.32} \tag{8}$$

$$P_t^{\mathrm{IT,opt}} = P_{\mathrm{idle}} + (P_{\mathrm{max}} - P_{\mathrm{idle}})(u_t)^{1.32} \tag{9}$$

$$E_t^{\mathrm{IT,base}} = P_t^{\mathrm{IT,base}}\Delta t, \qquad E_t^{\mathrm{IT,opt}} = P_t^{\mathrm{IT,opt}}\Delta t \tag{10}$$

---

$$E_{t+1}^{\mathrm{UPS}} = E_t^{\mathrm{UPS}} + \eta_{\mathrm{ch}} P_{\mathrm{ch},t}^{\mathrm{UPS}}\Delta t - \frac{P_{\mathrm{dis},t}^{\mathrm{UPS}}}{\eta_{\mathrm{dis}}}\Delta t \tag{11}$$

$$E_{\mathrm{min}}^{\mathrm{UPS}} \le E_t^{\mathrm{UPS}} \le E_{\mathrm{max}}^{\mathrm{UPS}} \tag{12}$$

$$E_{\mathrm{min}}^{\mathrm{UPS}} = SoC_{\mathrm{min}} E_{\mathrm{base}}^{\mathrm{UPS}}, \quad E_{\mathrm{max}}^{\mathrm{UPS}} = SoC_{\mathrm{max}} E_{\mathrm{base}}^{\mathrm{UPS}} \tag{13}$$

$$0 \le P_{\mathrm{ch},t}^{\mathrm{UPS}} \le z_t^{\mathrm{UPS}} P_{\mathrm{ch,max}}^{\mathrm{UPS}} \tag{14}$$

$$0 \le P_{\mathrm{dis},t}^{\mathrm{UPS}} \le (1 - z_t^{\mathrm{UPS}}) P_{\mathrm{dis,max}}^{\mathrm{UPS}} \tag{15}$$

$$P_t^{\mathrm{UPS,net}} = P_t^{\mathrm{UPS,ch}} - P_t^{\mathrm{UPS,dis}} \tag{16}$$

$$P_t^{\mathrm{IT}} = P_t^{\mathrm{grid,IT}} + P_{\mathrm{dis},t}^{\mathrm{UPS}} \tag{17}$$

---

The first power term adds the energy that reaches the battery after charging losses. The second removes more energy from the stored state than is delivered to the IT load, because discharge losses must also be supplied by the battery. The equation is applied from one interval to the next, so the operating choice made now changes the flexibility available later in the day and on subsequent days. The energy state is bounded by Equation (12); the limits are defined explicitly by Equation (13). For the central case, $SoC_{\mathrm{min}} = 0.5$ and $SoC_{\mathrm{max}} = 1$. The first annual interval begins at $E_{\mathrm{max}}^{\mathrm{UPS}}$; no equation forces the battery back to its opening energy at the end of each local day.

The lower bound preserves the minimum reserve assumed for the UPS and prevents the optimisation from using capacity that would not be released in practice. The upper bound prevents overcharging. Unlike the original selected day study, the annual formulation does not require the battery to return to an artificial daily target. Its closing energy is passed to the following day, where it remains available or must be replenished according to the subsequent price signal. A binary variable $z_t^{\mathrm{UPS}}$ selects the operating direction through Equations (14) and (15). When $z_t^{\mathrm{UPS}} = 1$, charging may be positive and discharge is forced to zero. When it is zero, the reverse is true. This exclusivity is required because simultaneous charging and discharging would waste energy and could otherwise be used by the mathematical model to create an artificial operating point.

For reporting, its signed net power is defined by Equation (16). Positive net power denotes charging from the grid and negative net power denotes discharge behind the meter. This identity is retained because it makes the component plots easy to interpret, although the optimisation itself uses the separate non-negative charge and discharge variables.

The charging limit is 270 kW. The battery has a larger discharge nameplate, but its useful discharge in this model cannot exceed the IT demand it supplies. UPS discharge is placed behind the meter through Equation (17). The UPS does not appear as an independent export from the facility. Its discharge replaces part of the grid electricity that would otherwise supply the IT equipment. Charging has the opposite effect and increases metered demand. This sign convention is also important when the component results are examined: a negative UPS contribution represents discharge behind the meter, not power exported through a separate market connection. This formulation is intentionally conventional. The novelty of the study does not depend on a new battery model; it depends on how the UPS is coordinated with workload timing and cooling, and on whether the resulting grid response can be sustained and recovered.

### C. Cooling System Model

At facility level, the cooling and TES formulation is collected below so that the thermal balance, storage constraints, chiller conversion, node dynamics and operating limits can be reviewed together. Here $Q_t^{\mathrm{cool}}$ is the gross thermal cooling supplied by the direct chiller and TES paths. The calibrated effective-airflow factor $\kappa$ accounts in aggregate for mixing, bypass and nonuniform rack-flow effects in the reduced-order thermal model. The cooling term acting on the four represented thermal nodes is therefore defined as $Q_t^{\mathrm{cool,eff}} \equiv \kappa Q_t^{\mathrm{cool}}$. This effective term is a model calibration; it does not assert that the remaining fraction is an independently measured cooling loss.

$$Q_t^{\mathrm{CA}} + Q_t^{\mathrm{HA}} + Q_t^{\mathrm{R}} + Q_t^{\mathrm{ITm}} = Q_t^{\mathrm{out}} + Q_t^{\mathrm{IT}} - Q_t^{\mathrm{cool,eff}} \tag{18}$$

$$E_{t+1}^{\mathrm{TES}} = E_t^{\mathrm{TES}} + \left( \eta_{\mathrm{ch}}^{\mathrm{TES}} Q_{\mathrm{ch},t}^{\mathrm{TES}} - \frac{Q_{\mathrm{dis},t}^{\mathrm{TES}}}{\eta_{\mathrm{dis}}^{\mathrm{TES}}} \right) \Delta t \tag{19}$$

$$E_{\min}^{\mathrm{TES}} \le E_t^{\mathrm{TES}} \le E_{\max}^{\mathrm{TES}} \tag{20}$$

$$0 \le Q_{\mathrm{ch},t}^{\mathrm{TES}} \le z_t^{\mathrm{TES}} Q_{\mathrm{ch,max}}^{\mathrm{TES}} \tag{21}$$

$$0 \le Q_{\mathrm{dis},t}^{\mathrm{TES}} \le (1 - z_t^{\mathrm{TES}}) Q_{\mathrm{dis,max}}^{\mathrm{TES}} \tag{22}$$

$$Q_t^{\mathrm{cool}} = \mathrm{COP}\, P_t^{\mathrm{ch,CRAC}} + Q_{\mathrm{dis},t}^{\mathrm{TES}} \tag{23}$$

$$Q_{\mathrm{ch},t}^{\mathrm{TES}} = \mathrm{COP}\, P_t^{\mathrm{ch,TES}} \tag{24}$$

$$P_t^{\mathrm{ch,CRAC}} + P_t^{\mathrm{ch,TES}} \le P_{\max}^{\mathrm{ch}} \tag{25}$$

$$T_t^{\mathrm{in}} = T_{t+1}^{\mathrm{HA}} - \frac{Q_t^{\mathrm{cool}}}{\dot{m} c_p} \tag{26}$$

$$T_{t+1}^{\mathrm{IT}} = T_t^{\mathrm{IT}} + \frac{\Delta t_s}{C_{\mathrm{IT}}} \left[ P_t^{\mathrm{IT}} - G_{\mathrm{conv}} (T_{t+1}^{\mathrm{IT}} - T_{t+1}^{\mathrm{R}}) \right] \tag{27}$$

$$T_{t+1}^{\mathrm{R}} = T_t^{\mathrm{R}} + \frac{\Delta t_s}{C_{\mathrm{R}}} \left[ \dot{m} \kappa c_p (T_{t+1}^{\mathrm{CA}} - T_{t+1}^{\mathrm{R}}) + G_{\mathrm{conv}} (T_{t+1}^{\mathrm{IT}} - T_{t+1}^{\mathrm{R}}) \right] \tag{28}$$

$$T_{t+1}^{\mathrm{CA}} = T_t^{\mathrm{CA}} + \frac{\Delta t_s}{C_{\mathrm{CA}}} \left[ \dot{m} \kappa c_p (T_t^{\mathrm{in}} - T_{t+1}^{\mathrm{CA}}) - G_{\mathrm{cold}} (T_{t+1}^{\mathrm{CA}} - T^{\mathrm{out}}) \right] \tag{29}$$

$$T_{t+1}^{\mathrm{HA}} = T_t^{\mathrm{HA}} + \frac{\Delta t_s}{C_{\mathrm{HA}}} \left[ \dot{m} \kappa c_p (T_{t+1}^{\mathrm{R}} - T_{t+1}^{\mathrm{HA}}) \right] \tag{30}$$

$$P_t^{\mathrm{IT}} \le Q_t^{\mathrm{cool}} \le (T_{t+1}^{\mathrm{HA}} - T_{\min}^{\mathrm{CA}}) \dot{m} c_p \tag{31}$$

$$T_{\min}^j \le T_t^j \le T_{\max}^j, \qquad j \in \{\mathrm{in}, \mathrm{IT}, \mathrm{R}, \mathrm{CA}, \mathrm{HA}\} \tag{32}$$

---

In this block, each $Q$ is expressed in kW, $Q_t^{\mathrm{IT}} = P_t^{\mathrm{IT}}$, and $Q_t^{\mathrm{cool,eff}} = \kappa Q_t^{\mathrm{cool}}$. The implemented form of this balance is given by the individual IT-equipment, rack, cold-aisle and hot-aisle state equations in the block. Writing the node equations separately preserves the thermal inertia of each component instead of replacing the facility with a single lumped temperature.

The cooling system is represented as a closed thermal circuit. The chiller can send cooling directly to the computer room air conditioning system, or it can produce cooling in advance and store it in the TES tank. Direct chiller output removes heat at the time electricity is consumed. Charging the TES increases current chiller demand but makes stored cooling available later. Discharging the TES allows part of the cooling requirement to be met without running the chiller at the same level. This separation between the production and use of cooling is the source of TES flexibility.

The 1000 kWh thermal energy store has charge and discharge limits of 300 kW thermal and efficiencies of 0.9 in both directions. Defining charging and discharging heat flows as $Q_{\mathrm{ch},t}^{\mathrm{TES}}$ and $Q_{\mathrm{dis},t}^{\mathrm{TES}}$, its state is given by Equation (19) in the consolidated block. Charging adds useful thermal energy after the charging loss has been applied, whereas supplying $Q_{\mathrm{dis},t}^{\mathrm{TES}}$ to the CRAC removes a larger amount from the store because of the discharge loss. As with the UPS, the TES state is carried between local days. The optimiser cannot empty the tank at the end of each day and assume that it is full again the following morning. Its stored energy obeys Equation (20), and the paired constraints in Equations (21) and (22) use a binary mode variable to prevent simultaneous charging and discharging. The mode constraint gives the store one physical direction in each interval. Without it, simultaneous charging and discharging could appear in a solution even though it provides no credible operational benefit. The energy and power limits describe different restrictions: capacity determines how much cooling can be moved through time, while the 300 kW ratings determine how quickly that capacity can be charged or released.

The chiller supplies the CRAC system directly and can also charge the TES. With a coefficient of performance $\mathrm{COP} = 5$, its two conversion relations are given by Equations (23) and (24). In Equation (23), direct chiller cooling and TES discharge combine to meet the cooling delivered to the data hall. Equation (24) describes the separate chiller output sent into storage. A COP of five means that one unit of chiller electricity produces five units of thermal cooling under the fixed central case assumption. The two electrical chiller loads share the 400 kW limit in Equation (25). This shared limit prevents the model from treating the CRAC and TES charging paths as if they were supplied by two independent chillers. A decision to charge the store can reduce the electrical capacity available for immediate cooling, and the optimiser must retain enough capacity to keep the data hall within its temperature limits.

The thermal model is adapted from the four node formulation of Cupelli *et al.* [35]. It describes the IT equipment, rack air, cold aisle and hot aisle, while the chiller and TES determine the inlet air temperature. The equations use consistent kW and kJ units. The inlet temperature is defined by Equation (26). The hot aisle air is cooled before it re-enters the cold aisle. The temperature drop is determined by the cooling supplied and by the heat capacity rate of the circulating air, $\dot{m}c_p$. Increasing cooling lowers the inlet temperature, subject to the physical limit introduced below. The four temperature states are updated with backward Euler in Equations (27)–(30). These equations describe the path followed by heat through the facility. The IT equipment equation assumes that its electrical demand becomes heat. Part of this heat is retained in the thermal mass of the equipment and the remainder passes to the rack air through the conductance $G_{\text{conv}}$. The rack equation then combines that heat transfer with the flow of conditioned air from the cold aisle. The factor $\kappa$ is a calibrated effective-airflow factor applied to the nominal flow participating in the reduced-order node exchange. It captures aggregate mixing, bypass and nonuniform rack-flow effects; it is not interpreted as a direct measurement of a separate cooling-loss stream.

The cold aisle receives the cooled inlet air and also exchanges heat with the outside environment through $G_{\text{cold}}$. The hot aisle receives the air leaving the racks. No corresponding external loss term is included in the hot aisle equation because the model assumes hot aisle containment. The four heat capacities determine how quickly the component temperatures respond. A large thermal capacity slows the change in temperature and allows cooling electricity to be moved for a time without immediately violating an operating bound. This is the thermal inertia used as a flexibility resource; it is not treated as a source of energy that can be used indefinitely.

Backward Euler is used because the air flow time constants make a 15 minute explicit update unstable. Unlike the earlier formulation, the end states in Equations (27)–(30) are constrained variables rather than free initial values. The cold aisle is held between 18 and 23°C, with wider physical bounds on the remaining states. The additional condition in Equation (31) prevents unvalidated accumulation of IT heat from being counted as a free flexibility resource and avoids overcooling the inlet air.

The lower inequality requires the model to remove at least the heat rate associated with IT operation. It was introduced after tests showed that broad temperature bounds alone could allow heat to accumulate in ways that had not been validated for sustained operation. The upper inequality limits the cooling that can be delivered before the inlet air would fall below the minimum cold aisle temperature. Together, the two bounds ensure that cooling flexibility comes from the timing of chiller and TES operation and from finite thermal inertia, rather than from ignoring heat or producing physically unusable cooling.

The temperature-domain constraints applied at every interval boundary are given by Equation (32). The cold aisle is limited to 18–23°C in both the benchmark and optimised annual cases. Wider physical ranges are applied to the remaining thermal states as listed in the implementation appendix.

$$P_t^{\text{grid}} = P_t^{\text{grid,IT}} + P_t^{\text{UPS,ch}} + P_t^{\text{ch,CRAC}} + P_t^{\text{ch,TES}} + P^{\text{aux}} \tag{33}$$

$$0 \le P_t^{\text{grid}} \le P_{\max}^{\text{grid}} \tag{34}$$

$$P_t^{\text{grid,IT}},\ P_t^{\text{UPS,ch}},\ P_t^{\text{ch,CRAC}},\ P_t^{\text{ch,TES}} \ge 0 \tag{35}$$

$$\min \sum_{t\in\mathcal{H}_d} \pi_t P_t^{\text{grid}} \Delta t/1000 \tag{36}$$

### D. Grid balance and objective

Grid import includes IT demand supplied from the grid, UPS charging, both chiller loads and a constant 53.095 kW auxiliary demand: Equation (33) is written at the facility meter. UPS discharge does not appear as a separate negative term because its effect has already been accounted for in $P_t^{\text{grid,IT}}$ through Equation (17). In contrast, UPS charging is an additional metered load. TES discharge also enters indirectly by reducing the direct chiller power needed to satisfy Equation (23). This accounting prevents the same storage action from being counted twice.

The model represents import only. Its electrical variables satisfy the import limit in Equation (34) and the non-negativity domain in Equation (35). The structural central-case limit is 1723.095 kW. This prevents the UPS from being interpreted as a separate exporting generator and makes the sign convention at the facility meter explicit.

For each daily solve, the objective is given by Equation (36), where $\mathcal{H}_d$ contains the committed day and its look ahead and $\pi_t$ is the signed Intermittent Market Reference Price (IMRP) in GBP/MWh [36]. Calendar year reporting includes only committed 2025 intervals, so the overlapping look ahead is never counted twice. Storage degradation, standing losses, network charges and terminal energy credits are not included in the central objective.

Because $\pi_t$ is signed, moving consumption from a high positive price interval to a low or negative price interval can reduce cost even when total electricity use rises. The objective represents participation in an energy market, not a complete retail electricity bill. A demand charge, connection constraint or degradation cost would change the relative value of high charging power and could discourage some of the peaks observed in the optimised result.

### E. 365-Day Rolling-Horizon Formulation

The model represents an aggregate DC with 1 MW of rated IT power. The purpose of the model is not to reproduce the placement of individual jobs or the airflow around individual racks. It is to retain the states that determine whether the facility can alter its grid demand without losing computational work or moving the thermal and storage consequences outside the study window. Those states are the unfinished flexible

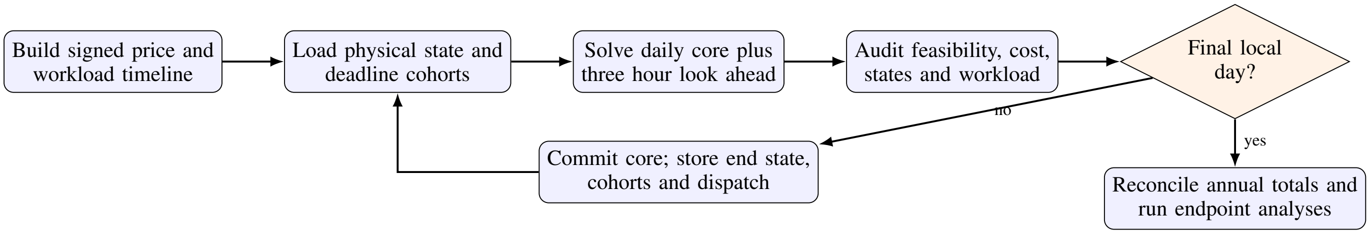



Fig. 4. Linked annual optimisation workflow. The stored checkpoint from one local day supplies the opening state of the next day and, later, the opening state of the representative day flexibility study.

workload, UPS and TES energy, the temperatures of the IT equipment, rack, cold aisle and hot aisle, and total grid import. This level of detail preserves the main interactions between computing, electrical storage and cooling while allowing the optimisation to be solved repeatedly over a complete year.

Time is divided into 15 minute intervals. Each solve contains one local calendar day followed by a 12 interval look ahead. A normal day therefore has 96 committed intervals, while the spring and autumn daylight saving transitions have 92 and 100. The additional three hours expose the latest deadline of every job that arrives during the committed day. Only the committed intervals enter the reported annual cost; the look ahead is used to protect workload completion and provide a feasible continuation.

At the end of local day $d$, the physical state

$$\boldsymbol{x}_d^{\text{end}} = (E^{\text{UPS}}, E^{\text{TES}}, T^{\text{IT}}, T^{\text{R}}, T^{\text{CA}}, T^{\text{HA}}) \tag{37}$$

is imposed as the opening state of day $d+1$. Unfinished workload is passed forward separately as a set of cohorts. Each cohort retains its original arrival time, its absolute deadline and its remaining CPU hours. This is important: resetting either the physical states or the workload at midnight would create energy or computing capacity that the DC does not actually possess.

Figure 4 shows the complete workflow. Each committed day is accompanied by a checkpoint and an audit record, so an interrupted annual run can resume from the last verified state. An absolute UTC timeline prevents missing or duplicated intervals, while local dates define the settlement day.

## IV. Case Study Design

The case study contains two connected experiments. The first follows economic operation through the full year and establishes the state from which flexibility would actually be offered. The second applies grid requests to one representative day taken from that annual result. Treating these as connected experiments avoids the optimistic assumption that every flexibility event begins with freshly reset storage and temperatures.

This sequence also separates two related questions. The annual experiment asks how the facility would operate when its only objective is to minimise signed electricity cost. The event experiment then asks what additional change in grid import remains feasible from that operating point. Flexibility is measured relative to a schedule the operator could actually be following, rather than relative to an arbitrary or deliberately favourable initial condition.

### A. Annual benchmark and optimised case

The annual run begins with 600 kWh in the UPS, 500 kWh in the TES and temperatures $(28.5, 26, 20, 27)$°C for the IT equipment, rack, cold aisle and hot aisle. No daily cyclic condition is imposed. The state at the end of one day becomes the state at the beginning of the next.

The benchmark executes flexible work at arrival and disables UPS and TES dispatch. Cooling remains cost optimised under the same thermal equations and temperature limits used in the optimised case. This gives a fair comparison of the value added by workload timing and storage without conflating that value with a different cooling model. The optimised case enables all three resources.

The benchmark is not intended to describe an unmanaged facility in every respect. It retains rational cooling operation because otherwise the comparison would attribute savings to a change in the cooling model itself. The difference between the two cases lies in the ability to move computational work and to coordinate the two stores. Both cases see the same workload arrivals, prices, weather assumption and physical temperature limits.

The price signal $\pi_t$ is the actual 2025 Intermittent Market Reference Price (IMRP) published by the Low Carbon Contracts Company [36]. The source is an hourly Great Britain day-ahead reference series. Each hourly observation is assigned unchanged to its four consecutive 15-minute model intervals; no interpolation or smoothing is applied, and negative prices are retained with their original sign. The final three-hour look-ahead beyond 31 December is supplied by the actual IMRP observations for 1 January 2026. Those intervals certify state and workload continuity but are excluded from the reported 2025 cost. IMRP is used here as a wholesale reference signal, not as a complete site retail tariff: network charges, levies, capacity charges and supplier margins are outside the objective.

The 53.095 kW auxiliary demand is a fixed modelling assumption rather than a measured profile or an optimisation decision. It is set to 7% of the average benchmark IT-plus-cooling demand (758.5 kW), giving $0.07 \times 758.5 = 53.095$ kW. This modest constant overhead represents lighting, networking, controls and other facility loads omitted from the explicit IT and chiller models. Holding it constant isolates the flexibility of the modelled resources; a site application should replace it with measured auxiliary demand.

Each annual scenario is solved in Pyomo using HiGHS. The requested relative gap is 0.1%, the daily time limit is 300 s and a feasible incumbent is normally accepted only when

Table IV
CENTRAL CASE PARAMETERS USED IN THE ANNUAL STUDY.

| Parameter | Value | Unit | Parameter | Value | Unit |
|---|---|---|---|---|---|
| Time interval | 0.25 | h | Look ahead | 3 | h |
| IT idle power | 166.7 | kW | IT rated power | 1000 | kW |
| UPS capacity | 600 | kWh | UPS minimum charge | 50 | % |
| UPS charge limit | 270 | kW | UPS efficiencies | 0.82/0.92 | – |
| TES capacity | 1000 | kWh-th | TES power limits | 300/300 | kW-th |
| TES efficiencies | 0.9/0.9 | – | Chiller COP | 5 | – |
| Chiller input limit | 400 | kW | Auxiliary demand | 53.095 | kW |
| Cold aisle range | 18–23 | °C | Outdoor temperature | 22 | °C |
| Flexible delay classes | 0.5/1/2/3 | h | Requested MILP gap | 0.1 | % |

the reported gap is at most 1%. The runs reported here used Python 3.12.0, Pyomo 6.10.1 and HiGHS 1.15.1 on Windows 11 with an eight core AMD Ryzen 7 9800X3D processor.

### B. Sensitivity design

The sensitivity analysis changes one resource at a time. Flexible workload share, UPS energy capacity and TES energy capacity are each set to 0.5, 0.75, 1.25 and 1.5 times their central values. Every setting is evaluated over the complete 2025 trajectory of 365 linked days (52 weeks plus one day), with the central 1.0 case shared across the three sweeps. Storage power ratings are held fixed so the capacity tests isolate energy duration rather than simultaneously changing power. Initial energy is kept at the same fraction of the revised capacity.

### C. Representative day flexibility assessment

The representative day is chosen before the flexibility sweep. For each eligible normal length day, the price mean, dispersion, extremes and negative price share are combined with mean and peak grid import, CPU utilisation, daily cost, and opening UPS, TES and cold aisle states. After standardisation against the annual distribution, the day with the smallest root mean square distance is selected. This procedure identifies 22 September 2025.

The exact 96 interval trajectory for that day and the following 12 intervals are loaded from the annual result. For an event beginning at $t_0$, the six physical states are fixed to their stored values. The aggregate CPU path before the event is also fixed. The annual export retains aggregate CPU rather than every internal cohort decision, so deadline level cohort allocation is reconstructed to match that path. No workload or state preconditioning is credited to the event.

The reconstruction assigns the executed flexible work back to feasible arrival cohorts while preserving the aggregate CPU trajectory. Its purpose is to recover the remaining deadlines that are needed during the event study. A job that has already used most of its permitted delay cannot be treated as newly flexible, whereas a recently arrived job may still have several admissible execution intervals. Retaining this distinction prevents the flexibility calculation from creating a fresh set of deferrable work at the event start.

For a requested signed deviation $\Delta P$, grid import during the event must satisfy

$$\left| P_t^{\text{grid}} - (P_t^{\text{grid,base}} + \Delta P) \right| \leq 0.1 \text{ kW}. \tag{38}$$

Negative values denote reduced import and therefore upward flexibility; positive values denote increased import and downward flexibility. Start times are sampled hourly. Reduced import requests range from $-500$ to $-100$ kW in 50 kW steps, while increased import requests are 25, 50 and 75 kW.

The search first extends a feasible event in one hour steps and then refines the first failed interval to 15 minute resolution. The maximum search duration is 12 hours or the end of the local day. A flexibility result is accepted only when the post event horizon also recovers its storage and thermal state. If $H$ is the terminal boundary, storage must satisfy

$$E_H^{\text{UPS}} \geq E_H^{\text{UPS,base}}, \qquad E_H^{\text{TES}} \geq E_H^{\text{TES,base}}, \tag{39}$$

and each temperature must satisfy

$$\left| T_H^j - T_H^{j,\text{base}} \right| \leq 0.05^\circ\text{C}, \quad j \in \{\text{IT}, \text{R}, \text{CA}, \text{HA}\}. \tag{40}$$

These recovery requirements are central to the meaning of the reported duration. Without them, the model could claim a long response by emptying the UPS or TES, moving temperatures to their limits, and leaving the deferred computational work for a later period. That would transfer the cost of the event outside the tested window rather than demonstrate a repeatable service. The terminal constraints require the principal energy and temperature states to return to at least the condition of the stored annual baseline, while workload completion is checked separately through the cohort deadlines. Each trial uses a 60 s screening limit. A timeout without a feasible incumbent is recorded as unresolved rather than infeasible. If the next 15 minute step remains unresolved, the last feasible duration is retained as a certified lower bound and marked with a plus sign in the heat map.

The event solves are not objective-free feasibility problems. Every trial minimises the same signed-price cost objective as the annual formulation, subject to the event target and recovery constraints. The component dispatch plotted for an accepted duration is therefore the minimum-cost accepted feasible incumbent. This secondary economic selection does not make the internal allocation unique: ties can exist, and a time-limited solve can return an accepted incumbent without proving that no equal- or lower-cost allocation exists. Duration certification is based on the audited feasibility of the accepted solution, not on uniqueness of its component decomposition.

The realised grid response is decomposed as

$$\begin{aligned} \Delta P^{\text{grid}} = \Delta P^{\text{IT}} + \Delta P^{\text{ch,CRAC}} + \Delta P^{\text{ch,TES}} \\ + \Delta (P_{\text{ch}}^{\text{UPS}} - P_{\text{dis}}^{\text{UPS}}). \end{aligned} \tag{41}$$

Table V
FULL-YEAR OPERATING-COST AND SENSITIVITY RESULTS.

| Case | Cost (GBP) | Saving (%) | Change (pp) | Energy (GWh) |
|---|---|---|---|---|
| Baseline | 550,504 | 0.000 | – | 6.718 |
| Flexible workload 0.5× | 531,117 | 3.522 | -1.528 | 6.782 |
| Flexible workload 0.75× | 526,707 | 4.323 | -0.727 | 6.787 |
| Flexible workload 1.0× | 522,707 | 5.049 | +0.000 | 6.794 |
| Flexible workload 1.25× | 519,157 | 5.694 | +0.645 | 6.801 |
| Flexible workload 1.5× | 516,058 | 6.257 | +1.208 | 6.807 |
| UPS capacity 0.5× | 525,374 | 4.565 | -0.484 | 6.771 |
| UPS capacity 0.75× | 524,001 | 4.814 | -0.235 | 6.783 |
| UPS capacity 1.25× | 521,411 | 5.285 | +0.236 | 6.805 |
| UPS capacity 1.5× | 520,222 | 5.501 | +0.452 | 6.815 |
| TES capacity 0.5× | 524,080 | 4.800 | -0.249 | 6.784 |
| TES capacity 0.75× | 523,327 | 4.937 | -0.113 | 6.789 |
| TES capacity 1.25× | 522,319 | 5.120 | +0.071 | 6.797 |
| TES capacity 1.5× | 522,057 | 5.167 | +0.118 | 6.799 |

This identity provides a transparent account of how the same external service can be delivered by different combinations of workload, cooling, TES and UPS action. Every event is checked for target tracking, recovery, workload conservation, objective reconciliation, mode exclusivity and component closure.

The signs in Equation (41) refer to changes from the annual baseline. For a reduced import request, a fall in IT or chiller demand and an increase in UPS discharge all contribute in the requested direction. Individual components may nevertheless move the other way. For example, deferred work may later be executed while the UPS discharges more strongly so that the metered target is still met. The decomposition records these compensating actions instead of assigning the whole response to whichever asset first moved.

## V. RESULTS AND DISCUSSION

### A. Annual cost and energy use

The workload at arrival benchmark costs GBP 550,503.59 in 2025. Enabling workload scheduling, UPS dispatch and TES operation reduces this to GBP 522,707.23. The resulting saving is GBP 27,796.37, or 5.049%. This annual percentage is lower than the saving obtained from the original selected day, which is expected: the annual run includes long periods with modest price variation as well as the more favourable days on which shifting has greater value. Table V gives the completed full year endpoints.

The annual comparison is more demanding than selecting a day with a pronounced price spread. Every local day is included, storage conditions are inherited from the previous day, and any unfinished work remains visible until its deadline. Days with a nearly flat price provide little reason to move demand and dilute the percentage saving achieved on more volatile days. The 5.049% result should be interpreted as the outcome of operating continuously through the observed 2025 price series, rather than as an extrapolation from a particularly favourable daily example.

The optimised case imports 6.794 GWh compared with 6.718 GWh in the benchmark. The financial saving is produced by moving demand to cheaper signed price intervals, not by reducing annual electricity use. Peak import rises from 974.9 to 1681.9 kW because deferred work and storage charging can coincide. This is an energy market result rather than a claim that the optimised profile is preferable under every tariff. A capacity charge or a tighter connection limit would penalise the higher peak and could materially alter the schedule.

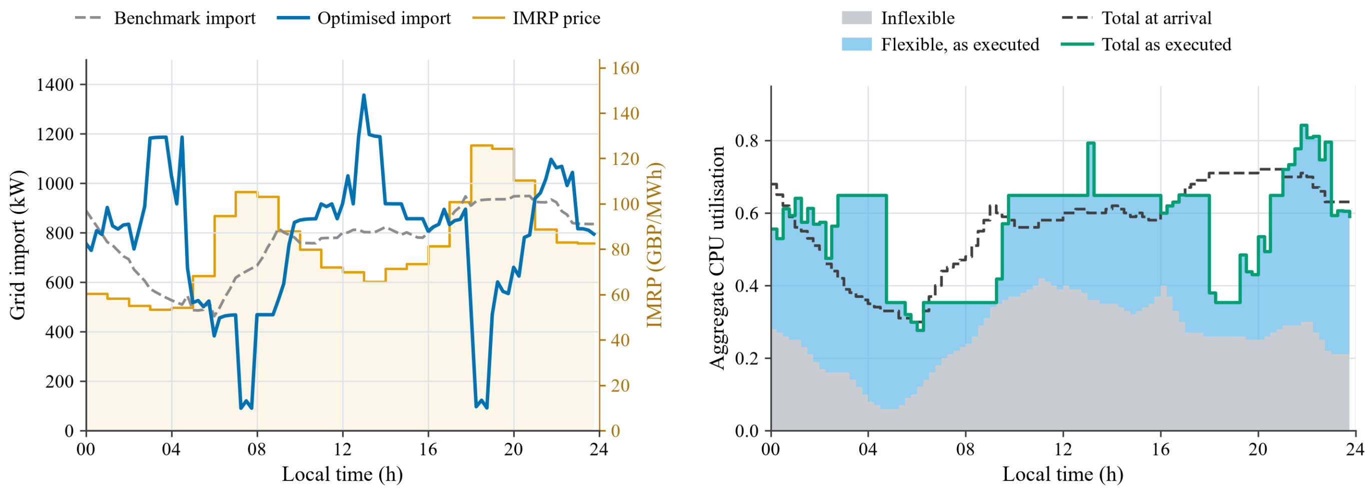
Table VI
ANNUAL SIGNED SETTLEMENT-COST COMPOSITION.

| Component | Baseline | Optimised (GBP) | Change |
|---|---|---|---|
| IT electrical demand | 407,653 | 392,414 | -15,239 |
| CRAC chiller | 105,337 | 91,251 | -14,085 |
| TES charging chiller | 0 | 6,905 | 6,905 |
| UPS net grid effect | 0 | -5,377 | -5,377 |
| Auxiliary overhead | 37,514 | 37,514 | 0 |
| Total settlement cost | 550,504 | 522,707 | -27,796 |

The increase in annual energy is about 1.13%, while the settlement cost falls by more than GBP 27,000. This apparent contrast is a consequence of two effects. First, charging and discharging the stores introduces losses. Second, the signed price objective can make additional consumption worthwhile when it occurs in a cheap or negatively priced interval and replaces demand at a more expensive time. The higher peak has a similar interpretation. It is economically admissible under the objective used here, but it would require explicit treatment before the result could be applied to a site with a firm import limit or a substantial capacity based tariff.

Table VI reconciles exactly to the settlement totals. Relative to the benchmark, the timing of IT demand reduces signed cost by GBP 15,239 and CRAC operation reduces it by GBP 14,085. TES charging adds GBP 6,905, while the net UPS grid effect reduces cost by GBP 5,377. The fixed auxiliary contribution is unchanged. These figures should not be read as standalone asset profits. Each component is part of the same jointly optimised schedule, and changing one resource changes the opportunities available to the others.

The positive TES charging entry does not mean that the TES has no value. It is the electricity cost assigned to producing stored cooling; the benefit appears through lower CRAC demand at other times and through the operating choices made possible for the rest of the system. In the same way, the CRAC saving cannot be attributed only to independent cooling




(a) Grid import and IMRP price.

(b) Workload execution.

Fig. 5. Annual benchmark and optimised operation on 22 September 2025. In (b), the shaded areas separate inflexible and flexible computation as executed; the dashed line shows the same day's workload at arrival.

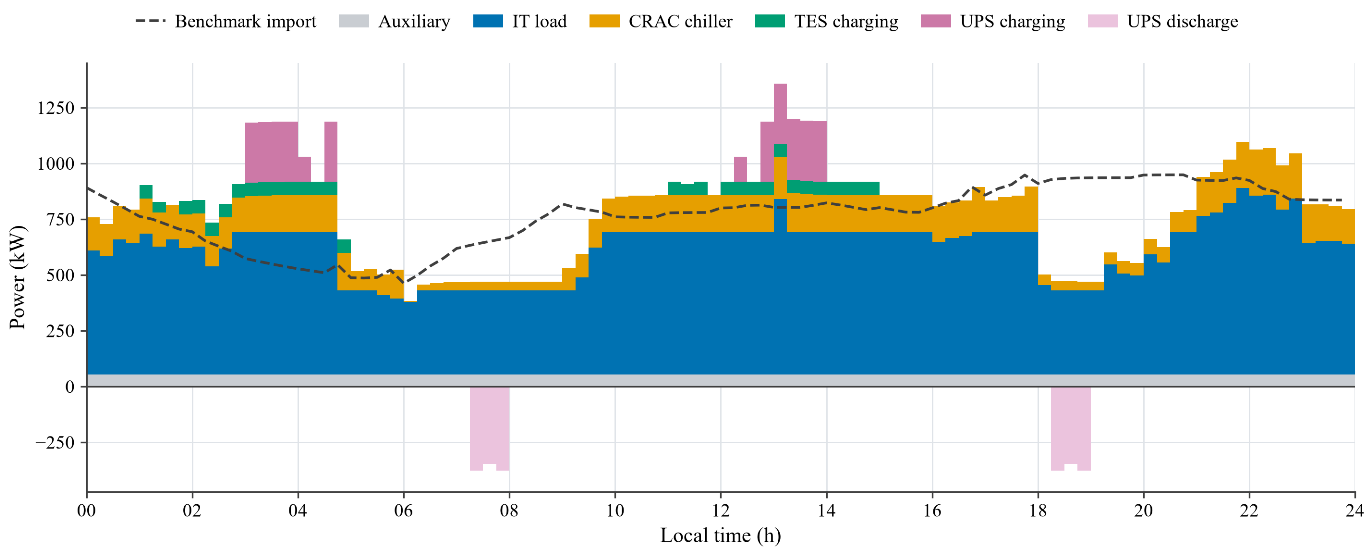



Fig. 6. Optimised component dispatch on 22 September 2025. Bars above the axis are metered demand. UPS discharge appears below the axis because it supplies IT demand behind the meter. The dashed line is benchmark grid import.

control, because moving IT work changes when heat enters the data hall. The table is best read as an exact accounting of one coordinated schedule, not as a set of asset business cases that can be added or removed without changing the other rows.

### *B. Operation on the representative day*

Figure 5 illustrates how flexible computation is moved away from expensive intervals and towards periods with lower or negative prices. The DC does not simply defer work to the end of the day. It uses the full set of permitted deadlines, so some jobs are moved by only one or two intervals while others use most of their available delay. The total amount of computational work is unchanged.

The two panels also show why the power trajectory cannot be inferred from price alone. The strongest reductions in optimised grid import occur around the morning and evening price increases, including the sharp troughs near 07:30 and 18:30. At other times, particularly around the middle of the day, import rises above the benchmark as work is completed and storage is charged. Some of this demand is preparation for later intervals and some is the consequence of jobs approaching their deadlines. The optimiser is balancing the current price against future workload obligations and the energy already held in the UPS and TES.

In Figure 5(b), the dashed arrival trajectory shows when computation would have been executed in the benchmark. The solid boundary and shaded areas show when it is actually executed. Periods in which the solid trajectory falls below the dashed line represent net deferral; later periods in which it rises above the dashed line include the completion of that deferred work. The distinction between flexible and inflexible utilisation makes clear that only part of the IT demand can respond. Even within the flexible part, the deadline classes prevent the optimiser from moving every job to the cheapest single interval.

Figure 6 shows the associated facility response. The UPS charges during selected low price periods and later supplies IT demand behind the meter. TES charging and CRAC demand are scheduled around both the price signal and the temperature constraints. The result is not a sequence of independent asset decisions: changing IT execution changes the heat input, which changes the cooling demand and the value of stored thermal energy. Both figures are slices through the annual trajectory. Their opening storage, temperature and workload conditions are inherited from 21 September rather than reset.

The component view makes these interactions more visible. In several overnight and midday intervals, metered demand exceeds the benchmark because IT execution, storage charging and cooling occur together. The UPS discharge below the axis near the morning and evening import troughs offsets part of the IT demand that remains at those times. TES charging is concentrated in selected intervals where chiller capacity and thermal headroom are available, rather than following price in isolation. CRAC electricity also changes when workload is rescheduled because the heat produced by the servers changes with IT power. Consequently, the same movement in grid import may involve a direct electrical response from the UPS, a delayed thermal response from the TES, and an indirect cooling response caused by workload timing.

This representative day is not a second optimisation with convenient opening conditions. It is a detailed view of one point in the completed annual chain. That distinction explains why an asset is not always charged or empty when the local price alone would suggest it should be: its opening state reflects the operation of 21 September, and its closing state must remain feasible for the following day.

### C. Sensitivity and asset ranking

Flexible workload is the most influential of the three tested resources. Halving its share reduces the annual saving to 3.522%, while increasing it by half raises the saving to 6.257%. UPS energy capacity ranks second, spanning 4.565–5.501%, and TES capacity ranks third, spanning 4.800–5.167%. Within the tested range, the ordering is flexible workload, UPS capacity and TES capacity.

The ranking is consistent with the physical role of the resources. IT demand is the largest variable load in the facility, and moving it also moves part of the cooling requirement. An increase in flexible share changes two parts of the grid demand trajectory and creates more opportunities throughout the year. UPS capacity has a smaller effect because its charge rate, efficiency and the IT demand available behind the meter limit how quickly additional energy can be used. TES capacity has the weakest endpoint response because its power rating is held fixed and its value depends on thermal headroom, chiller capacity and a later opportunity to displace direct cooling.

All three five-point annual curves are monotonic. Flexible workload and TES show diminishing marginal value, while the UPS response remains close to linear with a small taper at the upper end. The one-at-a-time design supports this resource ranking and response shape, but not a claim about interactions among simultaneous parameter changes; that would require a factorial sensitivity study.

### D. Flexibility duration and component use

Figure 7 reveals a strong dependence on time and direction. The longest response is a 25 kW increase in import beginning at 06:00, sustained for 9.75 hours. At the same start time, requests of 50 and 75 kW last 8.00 and 7.75 hours. Reduced import reaches 5.25 hours at 13:00 for requests of 100 and 150 kW, and also at 02:00 for 100 kW.

Each column of the heat map begins from a different point on the stored annual trajectory. The upper block contains increased import requests and the lower block contains reduced import requests. A darker cell indicates that the target can be maintained for longer from that start time. Blank cells are not missing data: they mean that no 15 minute response was certified at the requested magnitude. The plus signs distinguish a proven lower bound from an exact duration when the next candidate interval did not resolve within the screening limit.

The increased import results are strongest around the early morning. At these start times the facility can bring forward eligible work, increase the cooling associated with that work and charge storage while remaining within its later recovery requirements. Reduced import is more concentrated in the periods where the annual baseline leaves a combination of deferrable work, UPS energy and cooling headroom. The strong cluster around early afternoon is not a property of the installed capacities alone; it is a consequence of the operating state inherited on 22 September.

Duration does not decrease monotonically with magnitude at every start time. The request is an exact signed grid target in a mixed integer system, so changing the target can change which combination of workload deadlines, storage modes and thermal recovery remains reachable. This is one reason that a single power capacity value is an incomplete description of DC flexibility.

For example, at 15:00 the certified duration for a 200 kW reduction is 3.25 hours, whereas the 150 kW request is certified for only 0.75 hours. This does not imply that a larger reduction is generally easier. The exact targets activate different integer storage modes and workload schedules, and those schedules reach different recovery paths. A market product with a tolerance band rather than an exact target might smooth some of this behaviour, but the result demonstrates why magnitude and duration must be evaluated together.

Figures 8 and 9 show that similar grid requests can be produced by different internal actions. Reduced import combines IT deferral, lower CRAC or TES charging demand and UPS discharge. The deferred computation must still be completed, so compensating actions appear later in the event and recovery horizon. Increased import is produced mainly by advancing feasible work and increasing its associated cooling demand, with TES charging used when thermal headroom is available.

The reduced import panels illustrate how the available combination changes with start time. For the 06:00 requests

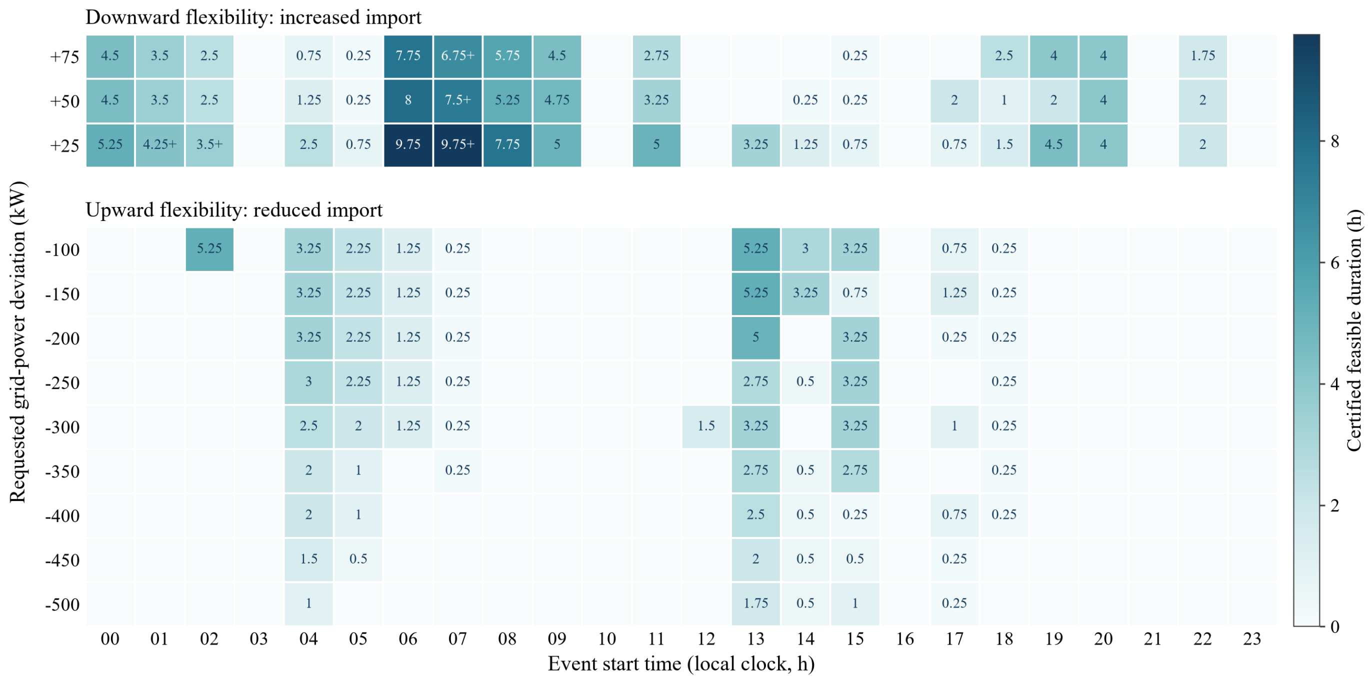



Fig. 7. Certified flexibility duration on 22 September 2025 for each start time and requested grid deviation. Blank cells have a certified duration of zero. A plus sign marks a lower bound where the next 15 minute step was unresolved at the screening limit.

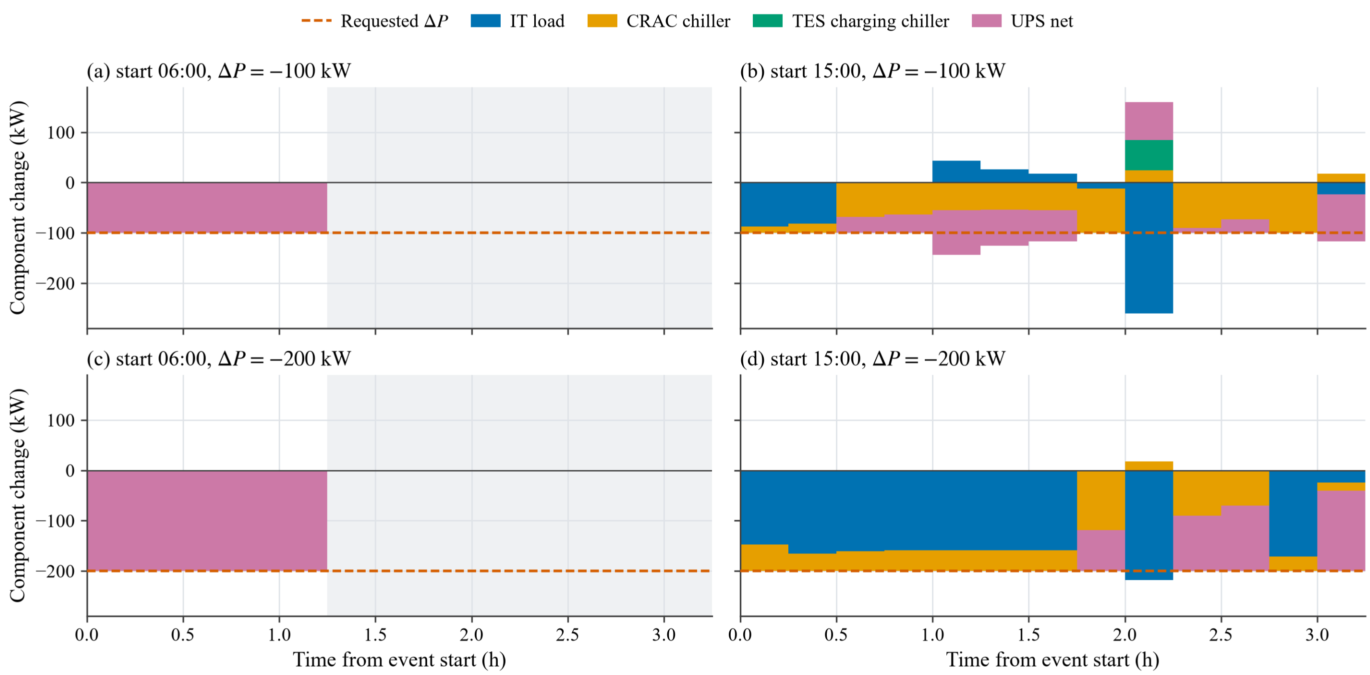



Fig. 8. Component decomposition for selected reduced import events. Positive and negative component changes are stacked separately and sum to the dashed requested deviation. The grey area begins after the certified event duration.

of 100 and 200 kW, UPS discharge supplies nearly all of the initial response. The grey recovery area begins after about 1.25 hours because the battery alone cannot sustain the exact target while also meeting the later state requirements. At 15:00, the response is distributed more widely. IT demand and direct cooling fall, UPS operation changes, and TES charging is adjusted. The 200 kW event then follows a different feasible path from the 150 kW heat map case described above.

The increased import panels show a different form of coordination. At midnight, the 25 and 75 kW requests are delivered through many relatively large internal changes that partly cancel at the meter. At 06:00, eligible work can be executed earlier than in the baseline and its associated cooling demand can rise, while UPS and TES actions manage the storage and temperature consequences. This permits the 25 kW event to continue for 9.75 hours and the 75 kW event for 7.75

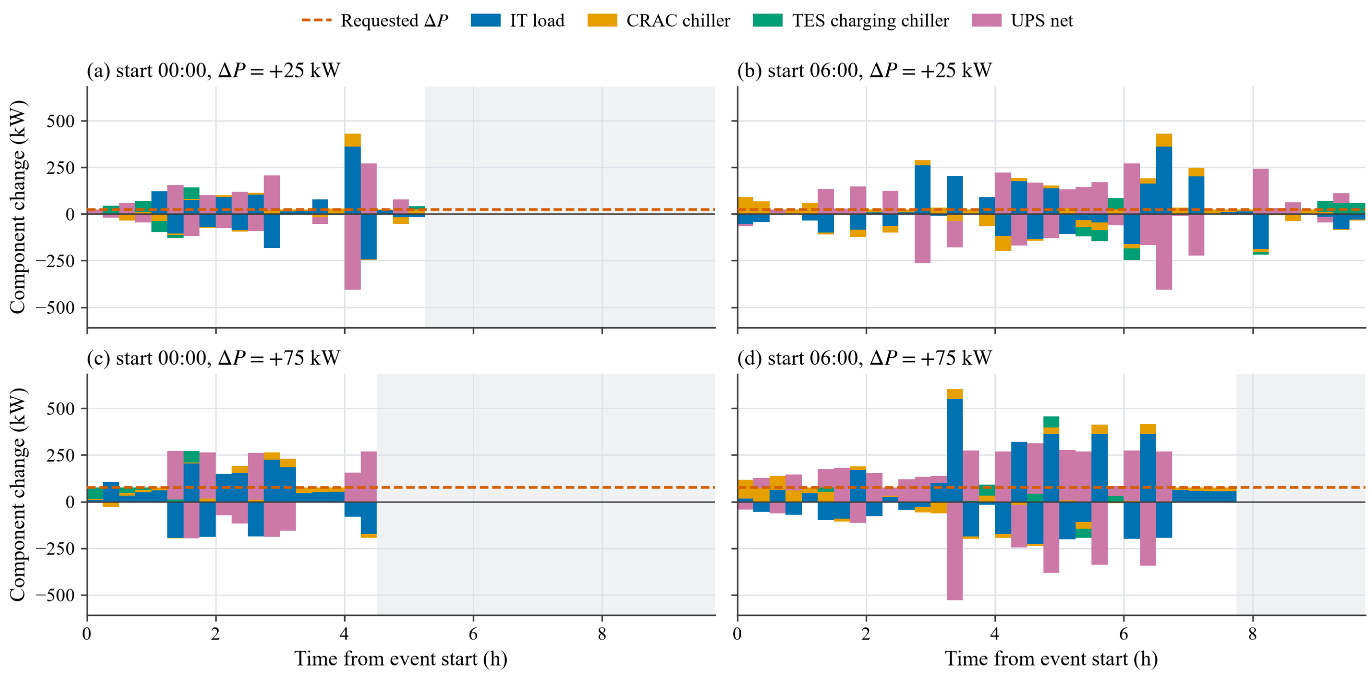



Fig. 9. Component decomposition for selected increased import events. Notation is as in Fig. 8.

hours. The long duration is not supplied by one asset holding a constant power level. It is assembled from a sequence of actions whose composition changes as deadlines and physical states evolve.

This interaction is precisely what separate asset models would miss. Deferring IT work changes cooling demand; TES affects when the chiller must run; and the UPS can support the grid while also helping the facility execute deferred work. The component bars close to the requested deviation to within $4.4 \times 10^{-11}$ kW. The largest event tracking error is 0.1 kW and the largest terminal temperature error is 0.05°C.

Some component bars are much larger than the requested change in grid import. This is not an accounting error. One resource may be restoring its state or executing deferred work while another moves further in the requested direction. Only their sum is visible to the grid. The small closure and tracking errors show that these opposing actions reconcile to the requested service at every interval.

### E. Continuity and solver quality

Every annual chain passes timestamp coverage, state handoff, workload conservation, charge and discharge exclusivity, and settlement reconciliation. The small number of time limited daily solves retain feasible incumbents. Exact-state 900 s reruns of the intermediate cases reduce the 25 May gap below 1% for the 0.75 UPS and TES cases; the 0.75 and 1.25 flexible-workload cases retain gaps of 1.015% and 2.032% because their negative objectives are close to zero. Across these checks the committed daily cost changes by at most GBP 0.281, the closing workload is unchanged and the maximum physical-state difference is $2.2 \times 10^{-14}$. The most conservative 0.5 flexible-workload endpoint has the same behaviour: its 900 s rerun changes committed cost by about GBP 0.002 despite a 5.652% relative gap. These qualifications do not affect any rounded annual percentage or the resource ranking.

At midnight after 31 December, 0.642875 CPU hours remain within deadline. They are completed in the actual 1 January 2026 look ahead. Calendar year reporting includes only 2025 committed intervals, while the extra intervals certify that the deferred work has not been abandoned. A continuation value test spanning the median to the maximum observed 1 January price changes annual cost by at most 0.000381%. This supports the zero terminal value used in the central case.

The flexibility grid contains 288 cells on a uniform hourly start time mesh. All accepted dispatches pass the physical audits and solver gap criterion. Exact duration boundaries are established for 283 cells. Five retain certified lower bounds because the next 15 minute increment did not resolve within the 60 s screening limit. Marking these cells explicitly avoids turning a solver timeout into an unsupported claim of infeasibility.

### F. Scope and limitations

This is a stylised operating study rather than a forecast for a named facility. Ambient temperature, chiller performance and the local clock workload shape are fixed. Battery degradation, storage standing losses, network tariffs and service revenue are omitted. These choices allow the annual value of coordination to be isolated, but they should be included before using the results for an investment case.

The repeated workload profile means that the annual variation in this study is driven mainly by prices and by the physical state carried between days, rather than by seasonal or weekly changes in computing demand. Fixed ambient temperature and COP similarly remove the seasonal variation in cooling efficiency. Depending on the facility and climate, introducing these effects could either create new opportunities or reduce the headroom available in warm periods. Battery losses are represented through efficiency, but degradation cost is not, so frequent UPS cycling may be less attractive in a commercial deployment than it is here.

The economic result also excludes network and capacity charges. This matters because the optimised annual case has a higher peak import. The 5.049% saving is evidence of value under the signed energy price objective, not a claim about the complete electricity bill of a particular site. Service revenue is not included either; the event study measures physical capability and does not test whether the service would be profitable after market payments and penalties.

The event study uses an exact grid target with a tolerance of 0.1 kW. This is a conservative representation and does not model the wider tolerance bands of a particular market product. The selected day gives a state consistent example of duration capability; it is not an annual availability distribution. Finally, the component contributions depend on the chosen dispatch and should not be interpreted as independently additive asset values.

Using one representative day gives a reproducible illustration of how the annual trajectory can support a grid event, but it does not show how often the same duration would be available across the year. Repeating the event study over a larger sample of seasonal and operational states would be needed before quoting an annual availability or dependable market capacity. The component results should likewise be interpreted as minimum-cost accepted feasible incumbents, not unique allocations of value among the assets.

## VI. Conclusion

This paper develops a linked annual optimisation for IT workload scheduling, UPS dispatch and cooling with thermal storage. Carrying physical states and unfinished workload between local days removes the artificial resets that can overstate both economic value and flexibility. For the 1 MW case study, coordinated operation reduces 2025 electricity cost from GBP 550,503.59 to GBP 522,707.23, a saving of 5.049%. Flexible workload has the largest effect among the tested resources, followed by UPS and TES energy capacity.

The annual cost decomposition shows that the largest gross reductions come from the timing of IT demand and CRAC operation. TES and UPS alter those opportunities rather than acting as isolated sources of value. On the representative day, the same stored annual trajectory is used to test grid service. Increased import can be sustained for 9.75 hours at 25 kW, while reduced import reaches 5.25 hours at 100 and 150 kW. The component results show how workload, cooling, TES and UPS actions combine differently with event direction, start time and the inherited operating state.

The main conclusion is not a single flexibility capacity. It is that DC flexibility has a time dependent magnitude and duration, and that a credible offer must include the recovery needed to return the facility to normal operation. Future work should extend the annual formulation to variable weather, battery degradation, network constraints, market revenue and multiple representative days, and should examine interactions among resources with a factorial sensitivity design.

## Appendix A
## Additional Implementation Detail

The DLOG representation of Equation (1) uses four segments with breakpoints

$$b_i = (i/4)^{1.5}, \qquad i = 0, \dots, 4, \tag{42}$$

and evaluates the power law exactly at each breakpoint. The logarithmic disaggregated convex combination requires fewer binary variables than a direct one binary per segment formulation while preserving equality to the piecewise curve.

State bounds are applied at every interval boundary, not only at the beginning and end of the daily solve. The UPS energy range is 300–600 kWh, the TES range is 0–1000 kWh thermal, and the IT, rack, cold aisle and hot aisle temperature ranges are 18–60, 18–40, 18–23 and 18–40°C. The inlet air is bounded between 18 and 30°C. The structural grid limit is the sum of rated IT power, maximum chiller input, auxiliary demand and maximum UPS charging, giving 1723.095 kW for the central case.

Each checkpoint records the committed dispatch, next physical state, unfinished cohorts, solver status and audit values. The annual audit tolerances are $10^{-5}$ for state handoff, $10^{-3}$ kW for power flow and $10^{-7}$ CPU hours for workload conservation. These records are also used to reconstruct the representative day without rerunning or resetting the annual trajectory.

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## Acknowledgements

This work was supported by the Engineering and Physical Sciences Research Council (EPSRC) and the Economic and Social Research Council (ESRC) through funding provided to the Energy Demand Research Centre (grant EP/Y010078/1).


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**James Day:** Writing – original draft, Investigation, Methodology, Conceptualization.

**Mehmet Türker Takci:** Writing – original draft, Investigation, Methodology, Conceptualization.

**Meysam Qadrdan:** Review & editing, Supervision, Conceptualization.